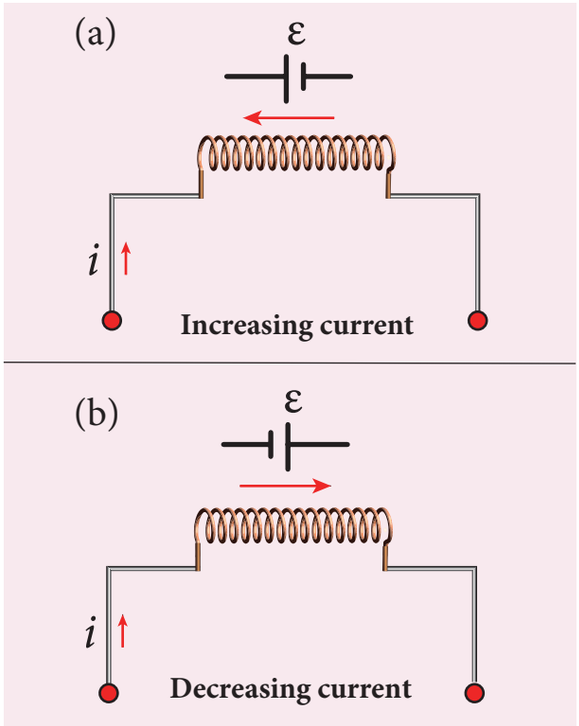


Figure 4.19 Induced emf ε opposes the changing current i

The inductance plays the same role in a circuit as mass and moment of inertia play in mechanical motion. When a circuit is switched on, the increasing current induces an emf which opposes the growth of current in a circuit (Figure 4.19(a)). Likewise, when circuit is broken, the decreasing current induces an emf in the reverse direction. This emf now opposes the decay of current (Figure 4.19(b)).

Thus, inductance of the coil opposes any change in current and tries to maintain the original state.

4.3.2 Self-inductance of a long solenoid

Consider a long solenoid of length l and cross-sectional area A. Let n be the number of turns per unit length (or turn density) of the solenoid. When an electric current i is passed through the solenoid, a magnetic

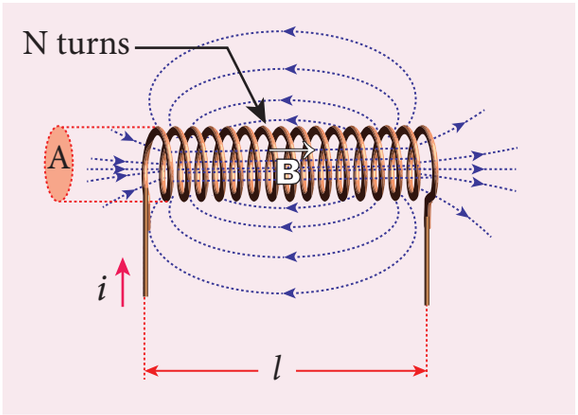


Figure 4.20 Self-inductance of a long solenoid

field produced inside is almost uniform and is directed along the axis of the solenoid as shown in Figure 4.20. The magnetic field at any point inside the solenoid is given by (Refer section 3.9.3)

$$B = \mu_{\circ} ni$$

As this magnetic field passes through the solenoid, the windings of the solenoid are linked by the field lines. The magnetic flux passing through each turn is

$$\Phi_B = \int_A \vec{B}.d\vec{A} = BA\cos\theta$$
$$= BA \qquad \text{since } \theta = 0°$$
$$= (\mu_0 ni)A$$

The total magnetic flux linked or flux linkage of the solenoid with N turns (the total number of turns N is given by $N = n\,l$) is

$$N\Phi_B = (nl)(\mu_0 ni)\,\mathrm{A}$$
$$N\Phi_B = (\mu_{\circ} n^2 Al)\,i$$

We know that

$$N\Phi_B = Li$$

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